Yiğit Sarp Avcı

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EDUCATION

Boğaziçi University — B.Sc. in Computer Engineering 09/2023 – Expected 06/2028

- Extracurriculars: Executive Board Member & DJ at Radyo Boğaziçi; Regional Organizational Team Member at ESTIEM; Work and Travel USA participant; Student-Athlete (sailing, skiing & windsurfing).
- Volunteering: Volunteering Division Member at Industrial Engineering Club (BÜYAK); Educational Mentor to high school students.

Kabataş Boys' High School 09/2018 – 06/2023

- GPA: 3.78/4.00 | Extracurriculars: Debate, Model United Nations, Entrepreneurship, Coding Clubs | Volunteering: CIP Club & TOÇEV.

EXPERIENCE

DefineX — Backend Developer Intern 09/2026 – Present

- Developing internal data processing platforms and Model Context Protocol (MCP) integrations, automating enterprise data extraction and RESTful backend pipelines.
- Leveraging modern AI-assisted engineering workflows to accelerate enterprise backend feature delivery, implementing robust validation and throughput optimization.

Akbank — Site Reliability Engineering Intern 07/2026 – 09/2026

- Maintained high availability for core banking infrastructure by designing real-time observability dashboards and proactive alerts in Dynatrace/Grafana.
- Engaged in Linux systems engineering, server health diagnostics, and log analysis to accelerate incident detection and root-cause resolution across production nodes.

Smash Mate — Software Engineering Intern 06/2026 – 07/2026

- Engineered player matchmaking, search discovery, and scoring microservices using Java (Quarkus), designing high-performance RESTful endpoints.
- Developed responsive mobile interfaces in React Native and authored comprehensive unit/integration test suites for pre-launch release.

TECHNICAL PROJECTS

Algorithmic Trading & Execution Engine Python, FastAPI, Redis, Next.js

- Engineered quantitative arbitrage engine achieving sub-second telemetry across 2s simulation loops with volatility-adjusted position sizing.
- Implemented a 4-state risk machine, Z-Score mean reversion, and real-time full-duplex WebSocket order book streaming via FastAPI and Redis.

Enterprise Operations & Knowledge Platform Next.js, Express 5, TypeORM, Jest

- Built full-stack workforce delegation and SOP platform boosting departmental auditing throughput across 5 core operational modules.
- Coupled an Express 5 REST API and TypeORM with SWR optimistic state caching, Recharts productivity analytics, and Supertest suites.

Restaurant Operations & Table Management SaaS Next.js, Zustand, Prisma, Zod

- Developed multi-tenant live-ops SaaS with slug isolation, eliminating seating bottlenecks via real-time floor plan coordinate sync.
- Engineered an interactive 2D canvas (react-rnd), waitstaff walk-in queues, and multi-table merging validated with Zustand and Zod.

Low-Level Systems & Architecture Lab x86-64, RISC-V, C11, Linux Syscalls

- Benchmarked ILP compiler flags (-O0 to -Ofast) with zero standard library overhead across Apple Silicon and RISC-V extensions.
- Authored raw x86-64 assembly syscall schedulers, bare-metal RV32I flight pathfinding on Spike (.ld memory maps), and a modular C11 Unix shell.

Linux IPC Game Subsystem C11, POSIX FIFOs, Pthreads, Signals

- Sustained deterministic 60 FPS frame rates and sub-millisecond controller response with zero process drops across game loops.
- Decoupled modular game execution via non-blocking FIFOs (O_NONBLOCK), POSIX signals, lock-free ring buffers, and dlopen hot-swapping.

Financial Analytics Dashboard FastAPI, Next.js, SQLAlchemy, OpenAI

- Delivered automated multi-asset diversification scoring and cash-flow analytics with 100% deterministic fallback reliability.
- Engineered a full-stack Next.js 14 and FastAPI platform with SQLAlchemy, PostgreSQL, and context-aware OpenAI advisory pipelines.

Cross-Platform & WebAssembly Game Lab C++17, Qt6, WASM, React Native

- Eliminated web execution latency and prevented mobile call-stack overflows during massive board state cascade clears.
- Compiled C++17/Qt6 2048 to WebAssembly via Emscripten with infinite undo, and built React Native Minesweeper with iterative BFS.

Advanced Data Structures & Algorithms Suite Java, Python, Algorithms, Graph Theory

- Resolved complex scheduling and graph challenges with optimal $O(V + E)$ linear time and $O(\log N)$ priority guarantees.
- Implemented Tarjan's SCC analyzer, indexed max-heaps, deterministic FSM engines, and an AST functional language interpreter.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C, C++, TypeScript, JavaScript, SQL, Assembly (x86-64, RISC-V)
- **Technologies & Frameworks:** Next.js, React Native, Quarkus, FastAPI, Express, Tailwind CSS, Prisma, TypeORM, PostgreSQL, Redis, Docker, WebSockets, WebAssembly, Qt6
- **Developer Tools:** Git/GitHub, Linux/POSIX, CI/CD (GitHub Actions), Jest, Pytest, Grafana, AI-Assisted Development
- **Languages:** Turkish (Native), English (Advanced), German (Beginner)
- **Hobbies:** Wakeboarding, Kiteboarding, Surfing, Bouldering, Running, Traveling